

PINE REPORT

(PANEL OF INFECTIOUS ENDOMETRITIS)

Pathogens Tested	Result
<i>Mycobacterium tuberculosis</i>	Not detected
Non-tuberculosis Mycobacterium species (NTM)	Not detected
<i>Neisseria gonorrhoeae</i>	Not detected
<i>Chlamydia trachomatis</i>	Not detected
<i>Mycoplasma genitalium</i>	Not detected
<i>Ureaplasma urealyticum</i>	Not detected
<i>Enterococcus faecalis</i>	Not detected
<i>Streptococcus agalactiae</i> (group B)	Not detected
<i>Staphylococcus aureus</i>	Not detected
<i>Escherichia coli</i>	Not detected
<i>Mycoplasma hominis</i>	Not detected
<i>Ureaplasma parvum</i>	Not detected

Interpretation and Recommendations:

- Dysbiosis as reflected by prevalence of < 90% of total lactobacillus species.
- Use *Lactobacillus crispatus* / *L. rhamnosus* vaginal probiotics to restore normal flora.
- The result can correlate with the following CST types (Community State Types) [1].
 - CST I – *L. crispatus* dominated, which is indicate most stable, healthiest profile.
 - CST II – *L. gasseri* dominated, which is indicating protective, less common.
 - CST III – *L. iners* dominated, which is indicating common, but less stable.
 - CST V – *L. jensenii* dominated, which is indicating rare, protective.
 - CST IV – Mixed diverse flora (low Lactobacillus, high anaerobes/pathogens → dysbiosis).

Test Description

The test was done to determine whether the uterine microbial environment is optimal for endometrial health. The method is based on detecting and measuring the amount of bacterial DNA present in the endometrial sample. Therefore, helps to determine when the endometrium presents a physiological bacterial flora.

Rapid Infection Detection, post initial library preparation and microbial DNA amplification, enables the detection of all known bacteria present in any clinical sample.

Test Methodology

DNA is extracted using QIAamp DNA Mini Kit. The extracted DNA is quantified using Qubit dsDNA High sensitivity Assay kit (Invitrogen, Cat# Q32854) (Qubit Fluorometer 3). The library is sequenced for 16S amplicon sequencing in SURFSeq 5000 platform.

After receiving the endometrial biopsy and extracting the genetic material (bacterial DNA), sample minimum quality requirements are evaluated before use of the diagnosis tools. The DNA sequences were mapped to the RefSeq version 91 (December 2018) and is analyzed using Internal pipeline. Minimum DNA volume used for Real time PCR to find 11 different pathogens (PINE).

The assay was performed using real time PCR based amplification of conserved gene specific regions of individual pathogens. Chronic endometritis is a persistent inflammation of the endometrial mucosa caused by bacterial pathogens [2].

The prevalence of chronic endometritis in infertile patients has been estimated at 2.8-39%, [3], [4], but can be as high as 60% or 66% in women diagnosed with unexplained recurrent miscarriage or repeated implantation failure, respectively [5],[6].

Test Limitation and Disclaimer

The aim of this test is to diagnosis patient's endometrial microbial health, including those pathogens causing chronic endometritis.

Factors such as hormonal changes, antibiotic intake, sexual activity, hygiene, etc may affect the results. Following the recommendations indicated in this report does not guarantee successful pregnancy. Failed implantation may be caused by other factors such as displaced window of implantation, poor embryo quality, genetic abnormalities, or previous pathologies.

This methodology is intended for the identification of bacteria at the genus level, although the species level can be defined in some cases. The correct identification of the species is clinically relevant for the detection of pathogens causing serious diseases as Mycobacterium tuberculosis, Chlamydia trachomatis, Neisseria gonorrhoeae. If DNA from those bacterial genus containing serious pathogenic species are found, we recommend to confirm the diagnosis with a specialist in infectious diseases. Also, this test is not intended for the diagnosis of sexually transmitted infections (STIs), if there is any suspect of these

types of diseases or the patient has a risk of suffer STIs, please consult with a specialist in infectious diseases to offer an appropriate clinical management. Detection limit of real time PCR (PINE) is 75 copies/ml. Below the detection limit will be reported as negative.

The results may be affected by contamination of the sample at the time of specimen collection and/or during transportation. Stringent precautions have been taken to prevent cross contamination in the laboratory.

References

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2. Molina, Nerea M., et al. "New opportunities for endometrial health by modifying uterine microbial composition: present or future?." *Biomolecules* 10.4 (2020): 593.
3. Inmaculada Moreno et al., The diagnosis of chronic endometritis in infertile asymptomatic women: a comparative study of histology, microbial cultures, hysteroscopy, and molecular microbiology. *AJOG*, JUNE 2018.
4. Johnston-MacAnanny E.B et al..Chronic endometritis is a frequent finding in women with recurrent implantation failure after in vitro fertilization. *Fertil Steril*. 2010.
5. Kasius J.C. Fatemi H.M. Bourgain C. et al. The impact of chronic endometritis on reproductive outcome. *Fertil Steril*. 2011.
6. Cicinelli E. Matteo M. Tinelli R. et al. Chronic endometritis due to common bacteria is prevalent in women with recurrent miscarriage as confirmed by improved pregnancy outcome after antibiotic treatment. *Reprod Sci*. 2014.
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Appendix

List of complete species level classification [click here](#)

	Report_Mrs.XXXXX	
Sample	Species	Abundance (%)
	<i>Lactobacillus crispatus</i>	18.46
XXXXX	<i>Lactobacillus sp.</i>	3.28
Mrs. XXXX	<i>Other</i>	78.26
	<i>Ralstonia pseudosolanacearum</i>	1.25
	<i>Paraburkholderia kururiensis</i>	1.22
	<i>Ralstonia solanacearum</i>	0.80
	<i>Paraburkholderia sp. SUR17</i>	0.76
	<i>Ralstonia syzygii</i>	0.67
	<i>Sphingobium vulgare</i>	0.60
	<i>Sphingobium xanthum</i>	0.56
	<i>Sphingobium sp. SYK-6</i>	0.48
	<i>Caulobacter segnis</i>	0.31
	<i>Novosphingobium fuchskuhlense</i>	0.31
	<i>Ralstonia mannitolilytica</i>	0.30
	<i>Aurantiacibacter suaedae</i>	0.28
	<i>Ralstonia pickettii</i>	0.26
	<i>Pedomonas mirosovicensis</i>	0.24
	<i>Novosphingobium capsulatum</i>	0.24
	<i>Caulobacter vibrioides</i>	0.21
	<i>Bradyrhizobium arachidis</i>	0.20
	<i>Novosphingobium naphthae</i>	0.19
	<i>Lactobacillus johnsonii</i>	0.19
	<i>Faucicola osloensis</i>	0.19

This report has been reviewed and approved by:



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